

Efficient Coverage Playbook

**Comparing Predictive Uncertainty Across Model Classes
for NFL Big Data Bowl Problems**

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NFL tracking data



frameId	x	y	s	a	dis	o	dir
7	67.68	29.89	0.34	1.57	0.04	124.86	88.21
8	67.76	29.89	0.69	2.13	0.08	124.07	89.59
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
50	73.67	25.06	4.19	2.62	0.42	134.21	125.26

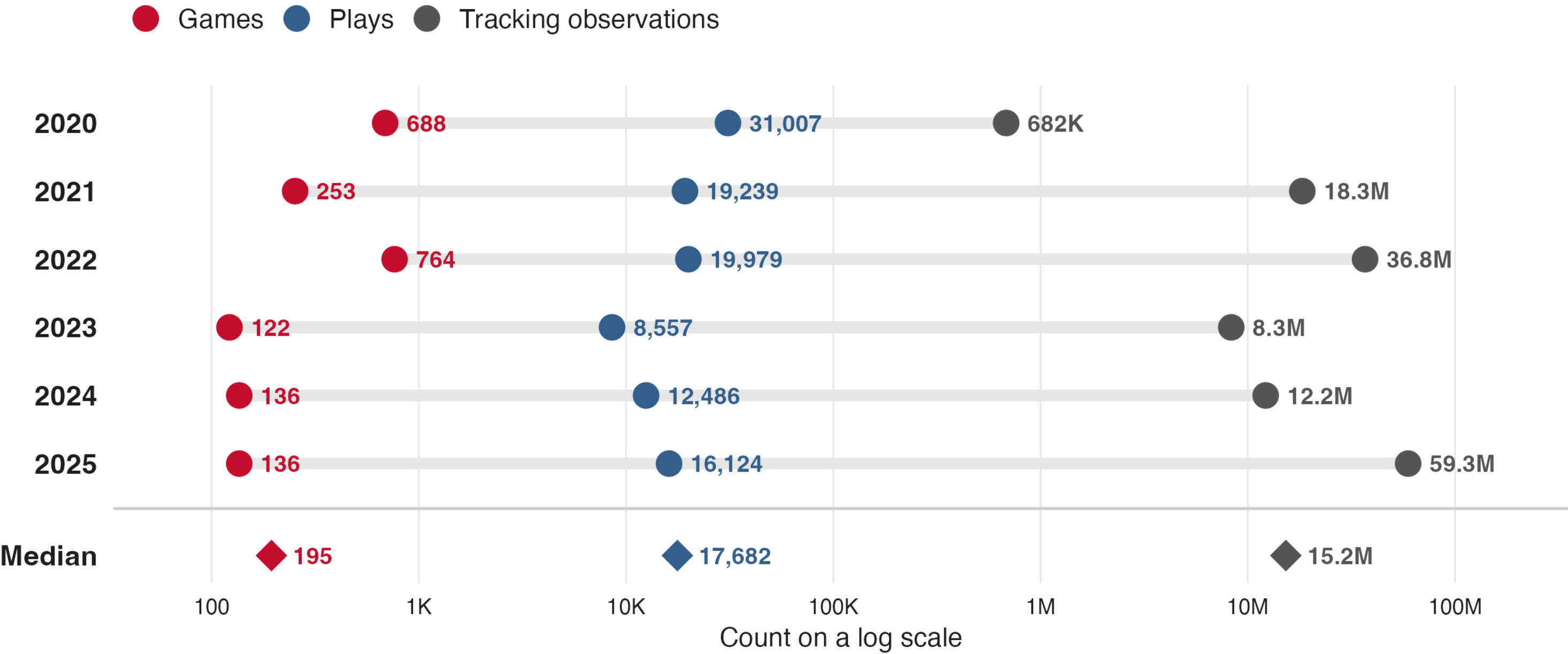


Player and football tracking features are reported at **10 Hz**.

Big Data Bowl problems, 2020–2025

YEAR	QUESTION	PREDICTION MOMENT	MODEL OUTPUT	ACCURACY
BDB2020	Rushing yards	Handoff	80-class distribution	CRPS
BDB2021	Pass completion	Pass release	Probability	Brier
BDB2022	Punt return yards	Punt received	Ordered distribution	CRPS
BDB2023	Sack	Snap	Probability	Brier
BDB2024	Tackle candidate	Candidate event –10 frames	Candidate probability	Brier
BDB2025	Man / Zone	20 frames through snap	Probability	Brier

The scale of Big Data Bowl datasets



BDB2020 has one handoff snapshot per rushing play. Later releases include 10 Hz tracking.

How do we evaluate predictive performance?

Accuracy

How closely do predictions match outcomes?

Proper loss functions compare predicted probabilities or distributions with observed outcomes.

Uncertainty

What outcomes remain plausible for this play?

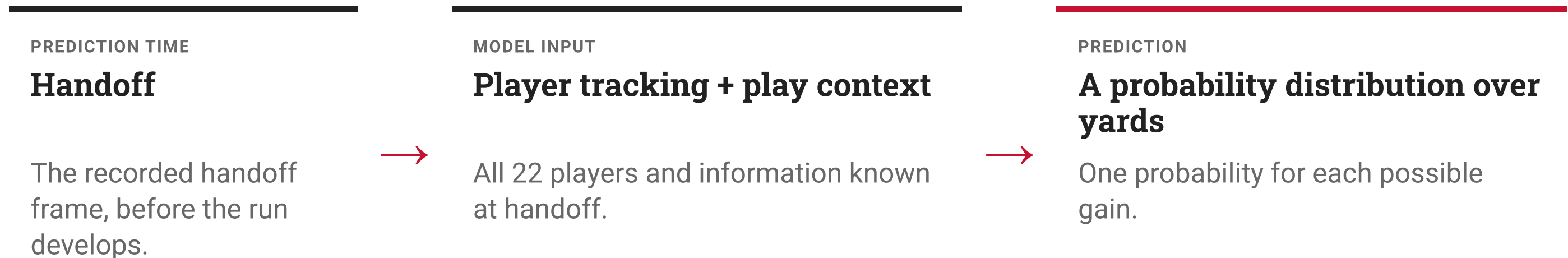
Prediction intervals or sets quantify uncertainty for a future outcome.

Stability

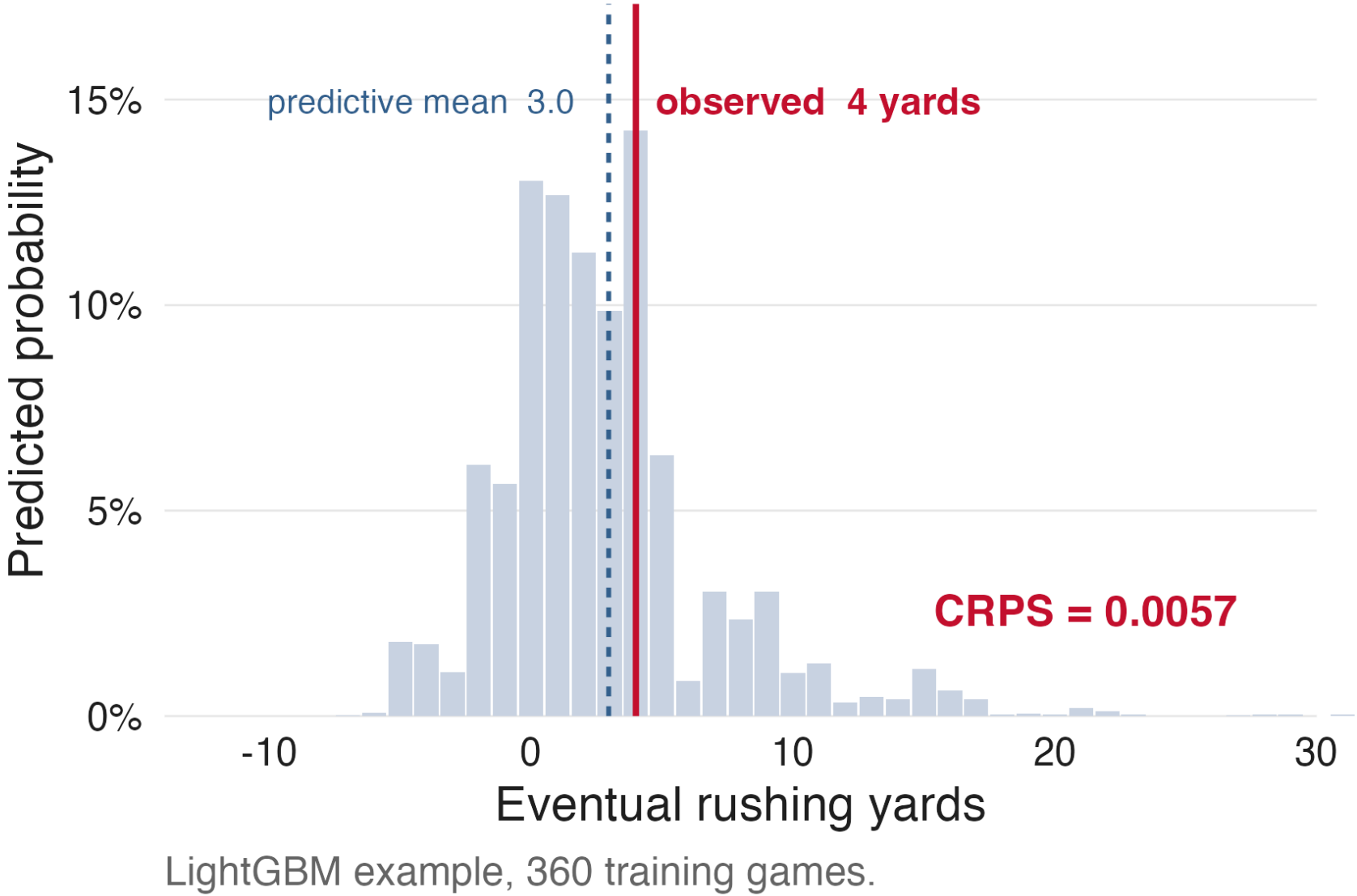
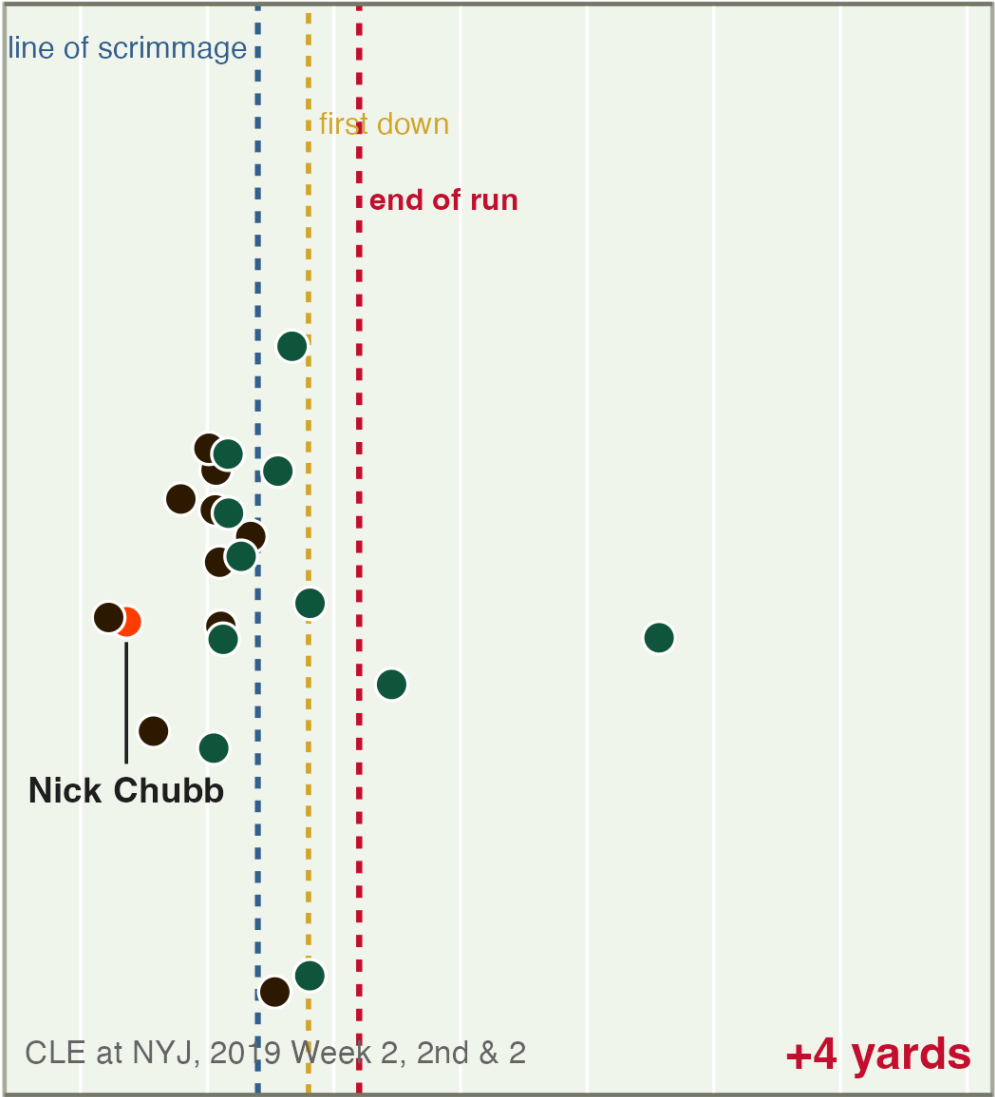
Would another sample of games change model estimates or rankings?

Repeated game splits measure variation in predictive performance and model rankings.

What is the distribution of a runner's eventual yardage?



A rushing yard prediction for one play



CRPS scores the full predicted distribution. Lower values indicate better predictions.

Model features at handoff

Engineered football features

Positions, motion, distances, and game context summarized into one table.

Used by regularized regression and boosted trees.

Tradeoff: compact and easier to fit, but the analyst chooses the football structure in advance.

Raw player tracking

Each player's position and movement at handoff.

Used by relational and attention models.

Tradeoff: preserves more detail, but the model must learn more structure from the same games.

Model classes across football problems

MODEL CLASS	INPUT	ESTIMATOR
Regularized regression	Engineered football features	Penalized linear or logistic model
Boosted trees	The same engineered features	Gradient boosted decision trees
Relational network	Tracking + a football graph	Shared pairwise MLP with averaged messages
Attention relational network	The same tracking and graph	Shared pairwise MLP with learned message weights
Sumer Transformer	Tracking without a specified graph	Attention across players and observed frames

The same five model classes are fit for every Big Data Bowl problem.

Split conformal prediction intervals

Leverage **calibration games** to determine intervals without refitting.

01

Fit the model

Fit on training games, then keep the prediction model fixed.

02

Score held-out plays

Use separate calibration games to score disagreement between predictions and outcomes.

03

Set a threshold

Take a high quantile of the calibration scores for the desired coverage.

04

Form the interval

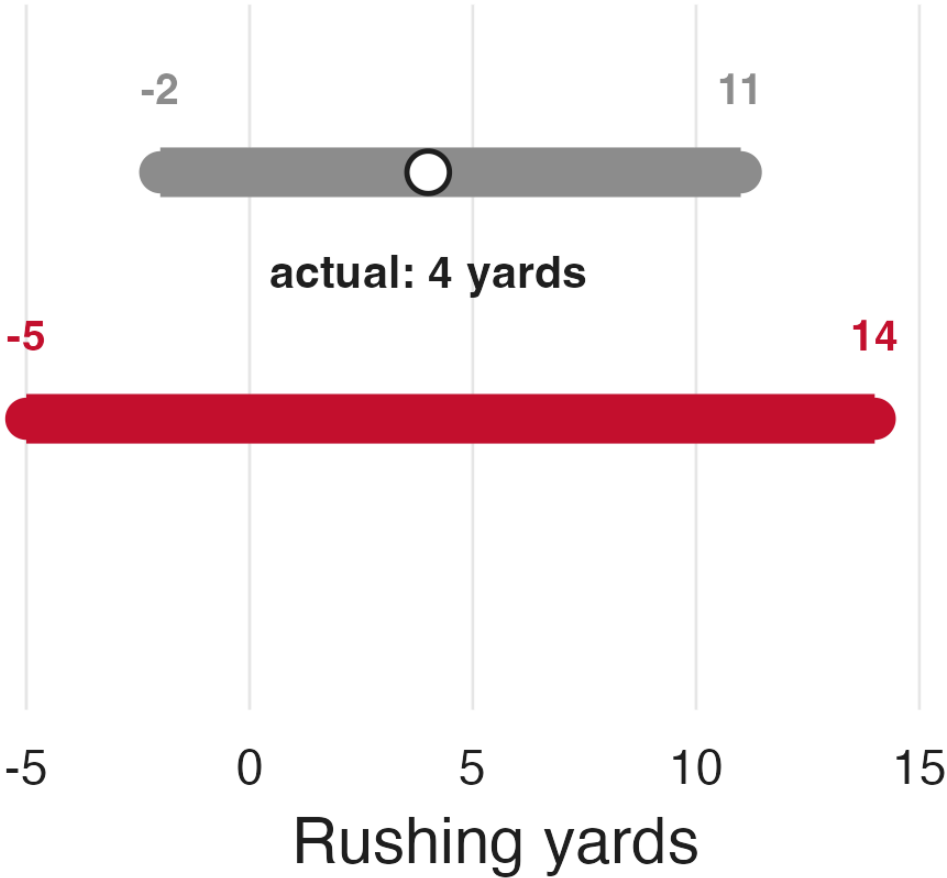
For a new play, include outcomes whose scores do not exceed the threshold.

Split conformal intervals for rushing yards

Example play: Nick Chubb vs. the Jets

Model's central interval

Example padding
(+3 yards)



Calibration

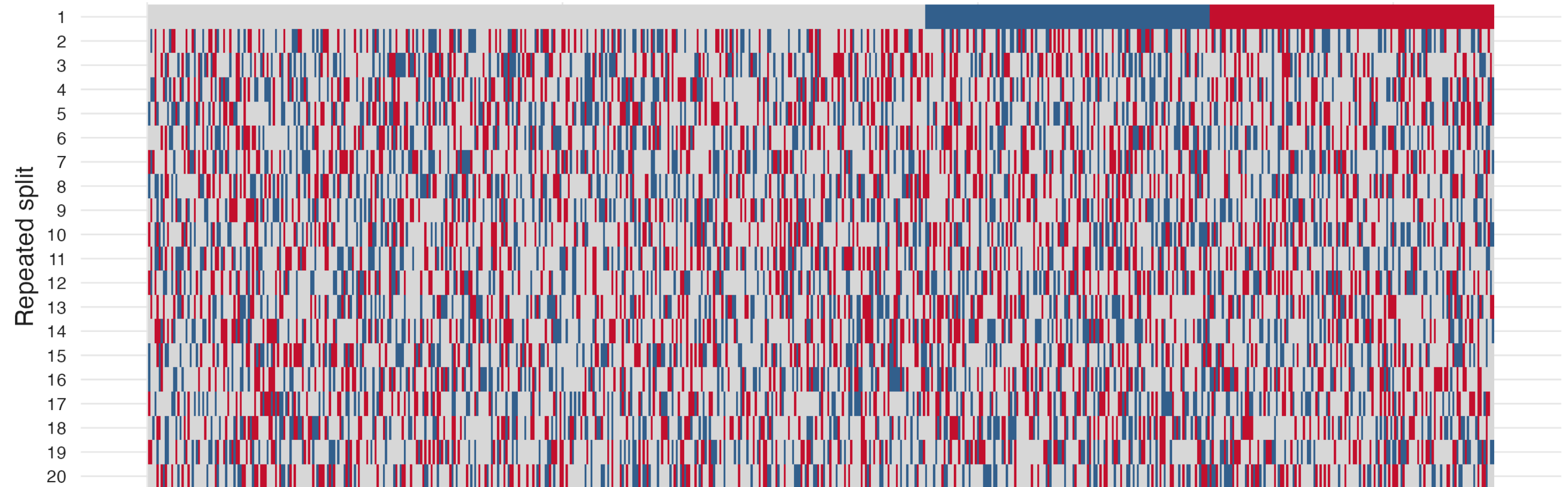
For one eligible play per held-out game, record the yards beyond the central interval (0 inside it).

Padding

The finite-sample quantile for 90% coverage gives the allowance added to both endpoints.

Game splits and predictive stability

We reshuffle the games in each repetition and fit every model at six training sample sizes.



The same 648 confirmatory games, reassigned each time

■ Training pool ■ Calibration ■ Test

First 20 of 100 repetitions shown. Models share each split. 40 development games stay fixed.

Comparing models across repetitions

Paired loss differences

Models A and B use the same test games in repetition r . Write their mean test losses as $\bar{L}_{A,r}$ and $\bar{L}_{B,r}$.

$$d_r = \bar{L}_{A,r} - \bar{L}_{B,r}$$

A negative difference favors A .

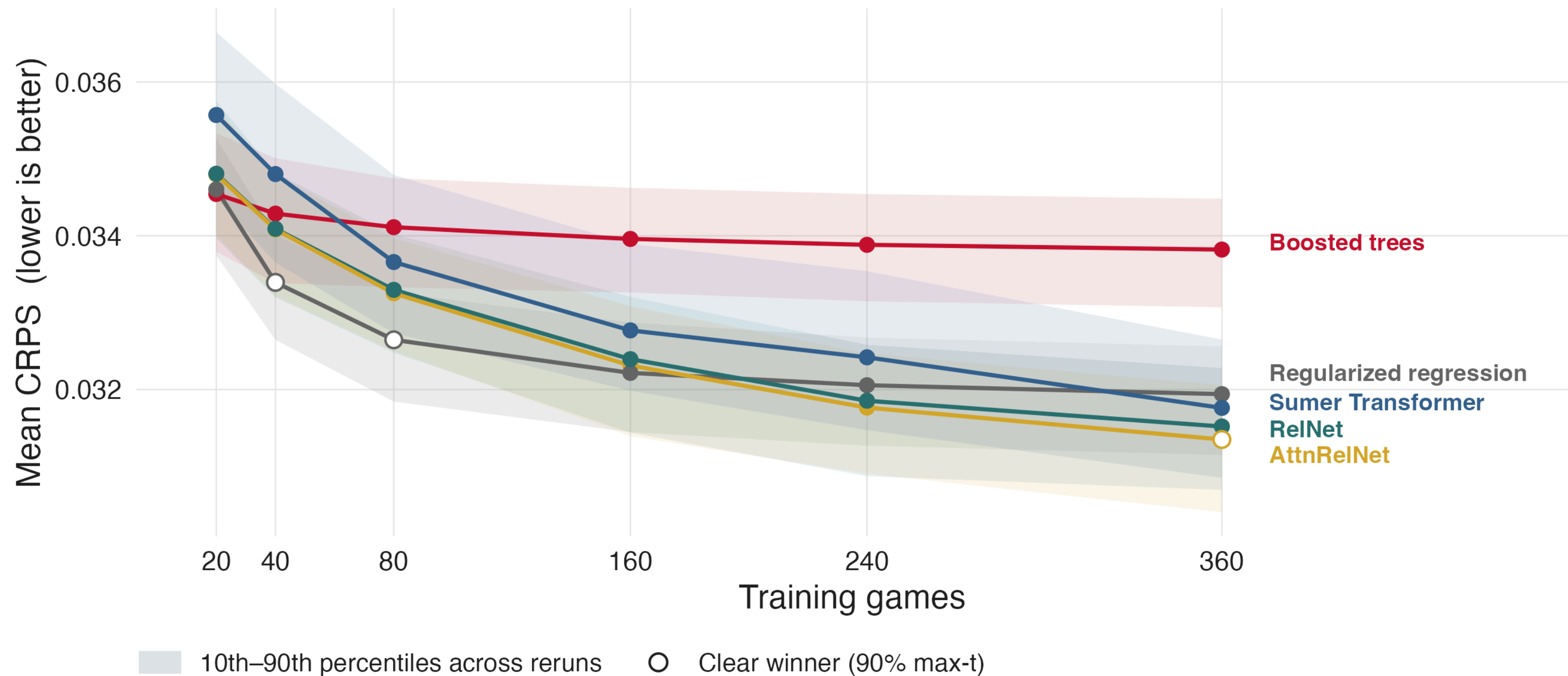
Simultaneous comparisons

Compare every pair of models at each training sample size: **10 model pairs × 6 sample sizes = 60 comparisons**.

Resample the 100 repetitions jointly. The **90% max-t intervals** account for dependence across all 60 comparisons within each problem.

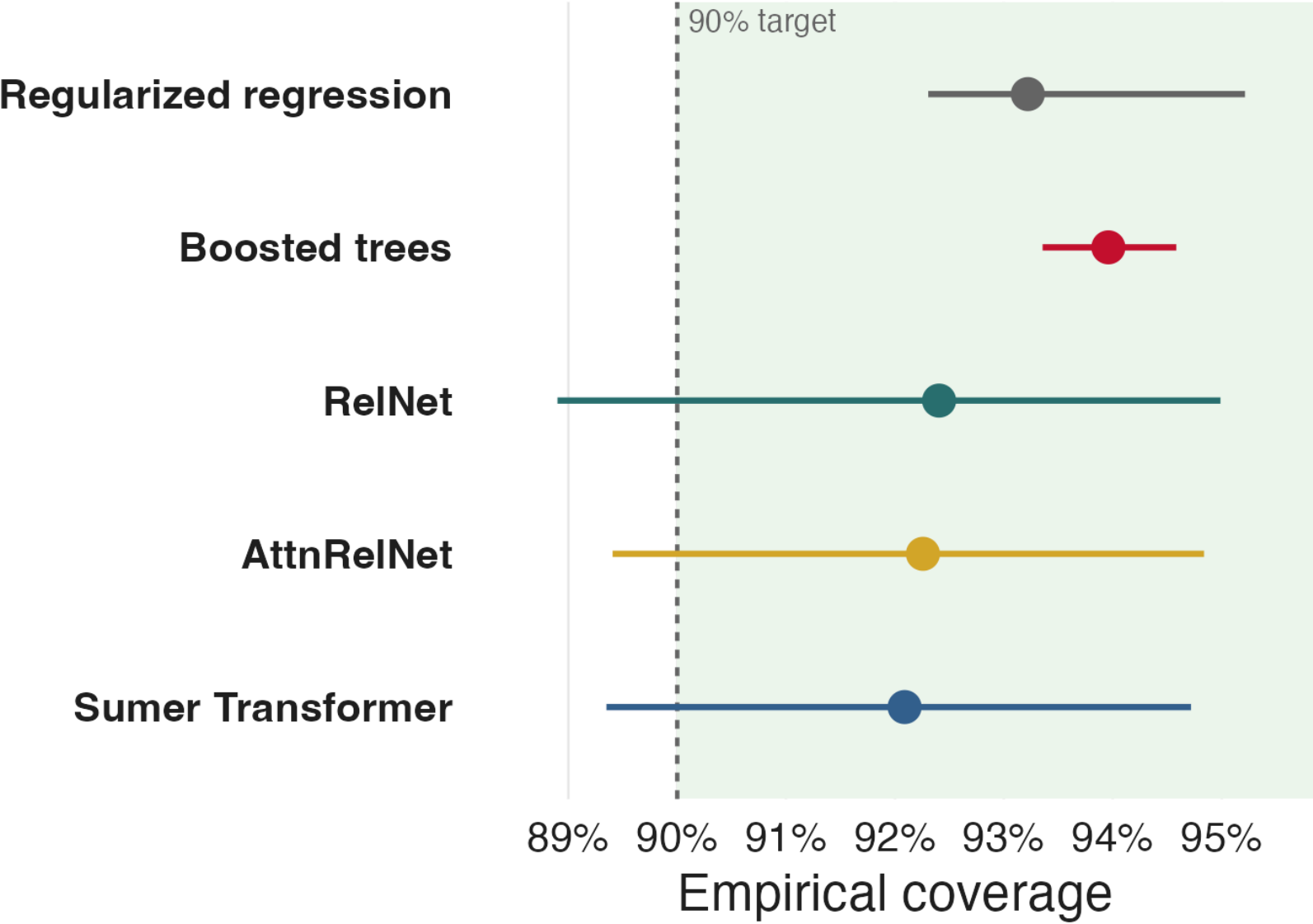
Clear winner: lower loss than all four alternatives at that training sample size under the paired 90% max-t intervals.

Rushing accuracy across sample sizes

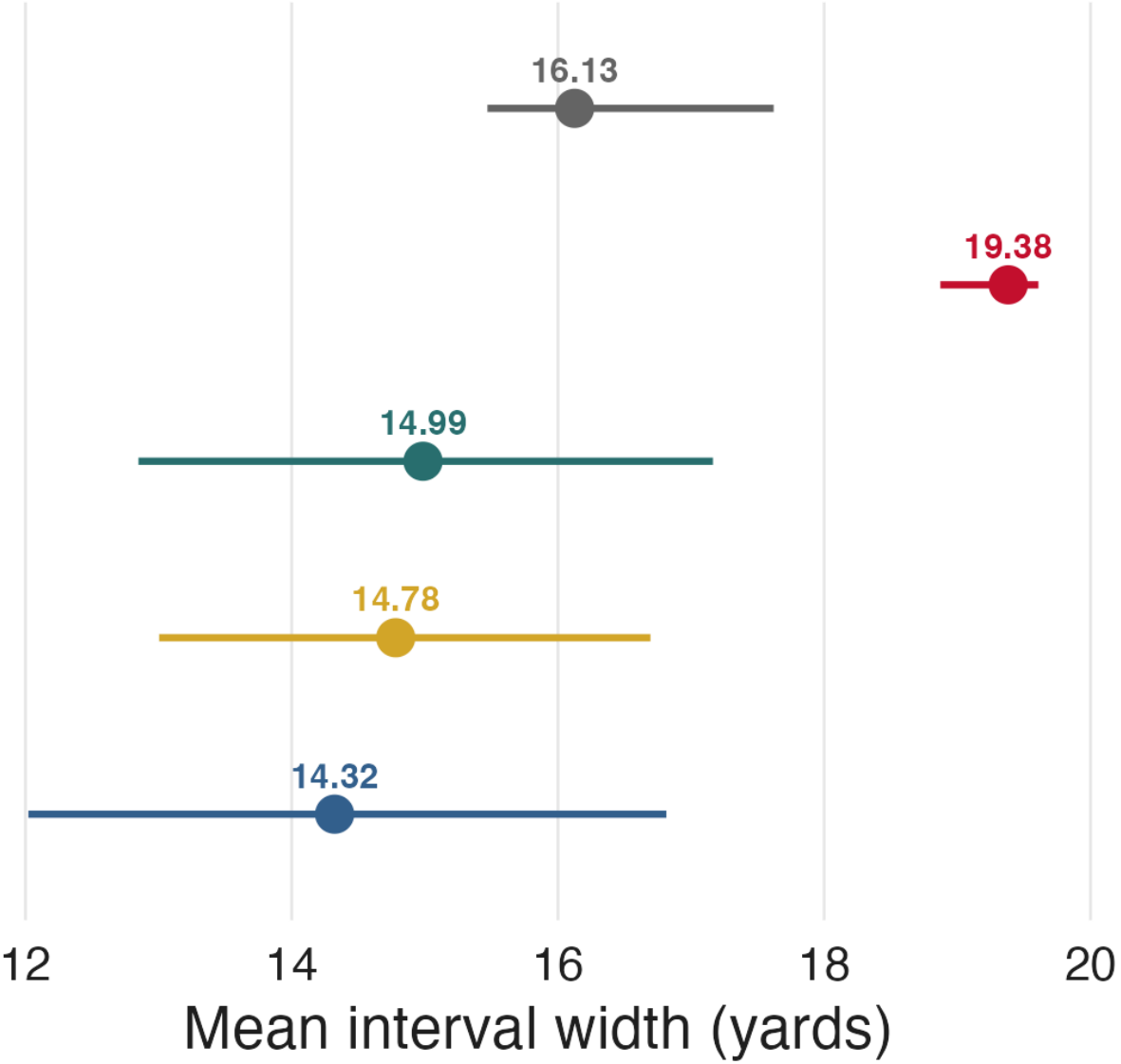


Rushing interval coverage and width

Coverage across repeated splits



Width across repeated splits



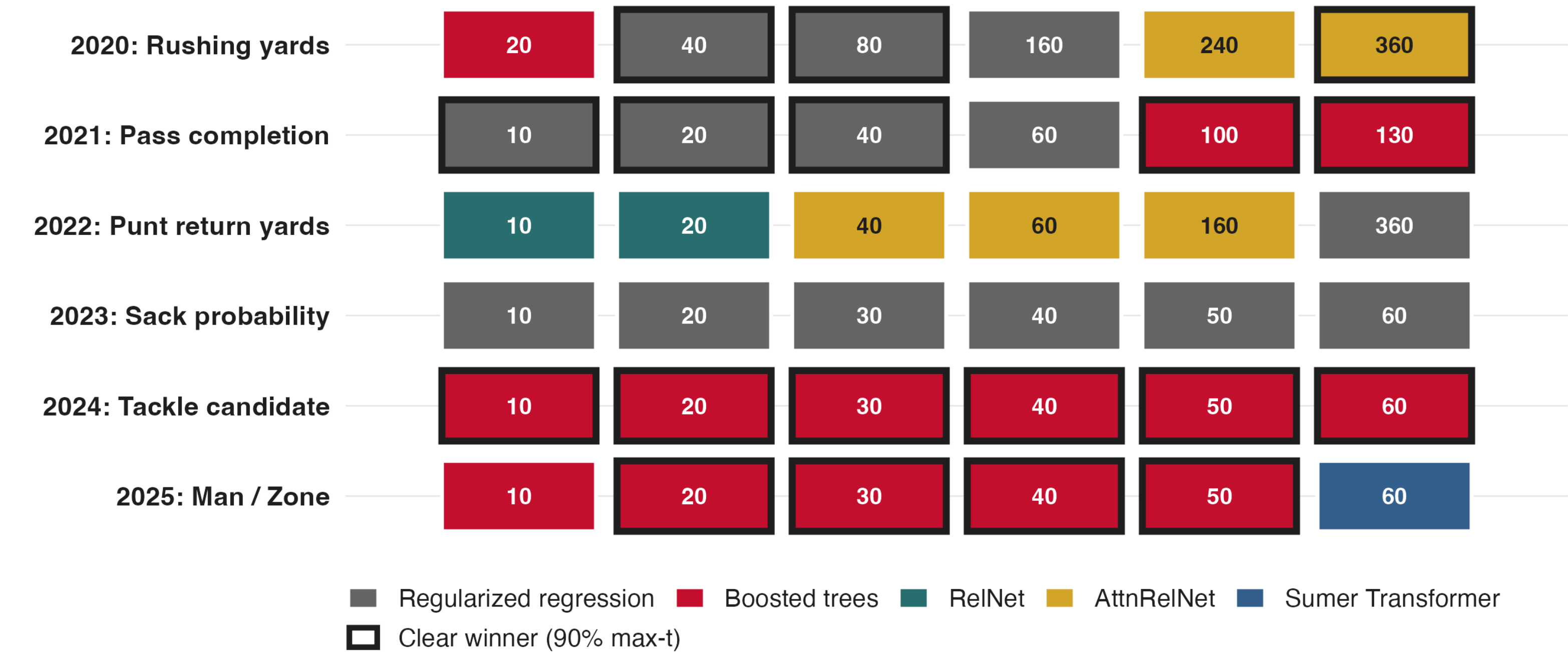
360 training games. Horizontal bars show 10th–90th percentiles across 100 repetitions.

BDB2020 model rankings by training sample size



Dark outline: clear winner under the problem's paired 90% max-t intervals.

Model rankings across football problems



Limitations and next steps

Across these problems, we only evaluate predictions at a single selected moment.

- **An illustrative extension:** instead of predicting rushing yards only at handoff, update the forecast throughout the run.
- **Accuracy:** how should performance be evaluated across repeated predictions?
- **Uncertainty:** coverage at one time does not guarantee coverage throughout the play. Protecting the full sequence may require wider intervals.

Conclusions

How do we evaluate the quality of a predictive model?

- **Accuracy is only one aspect of quality.** We also care about the coverage and informativeness of prediction intervals (sets), as well as variability across samples.
- The “**best**” model class depends on the problem, the number of training games, and the evaluation criterion.
- With limited sample, the practical question is how much flexibility the available data can support.

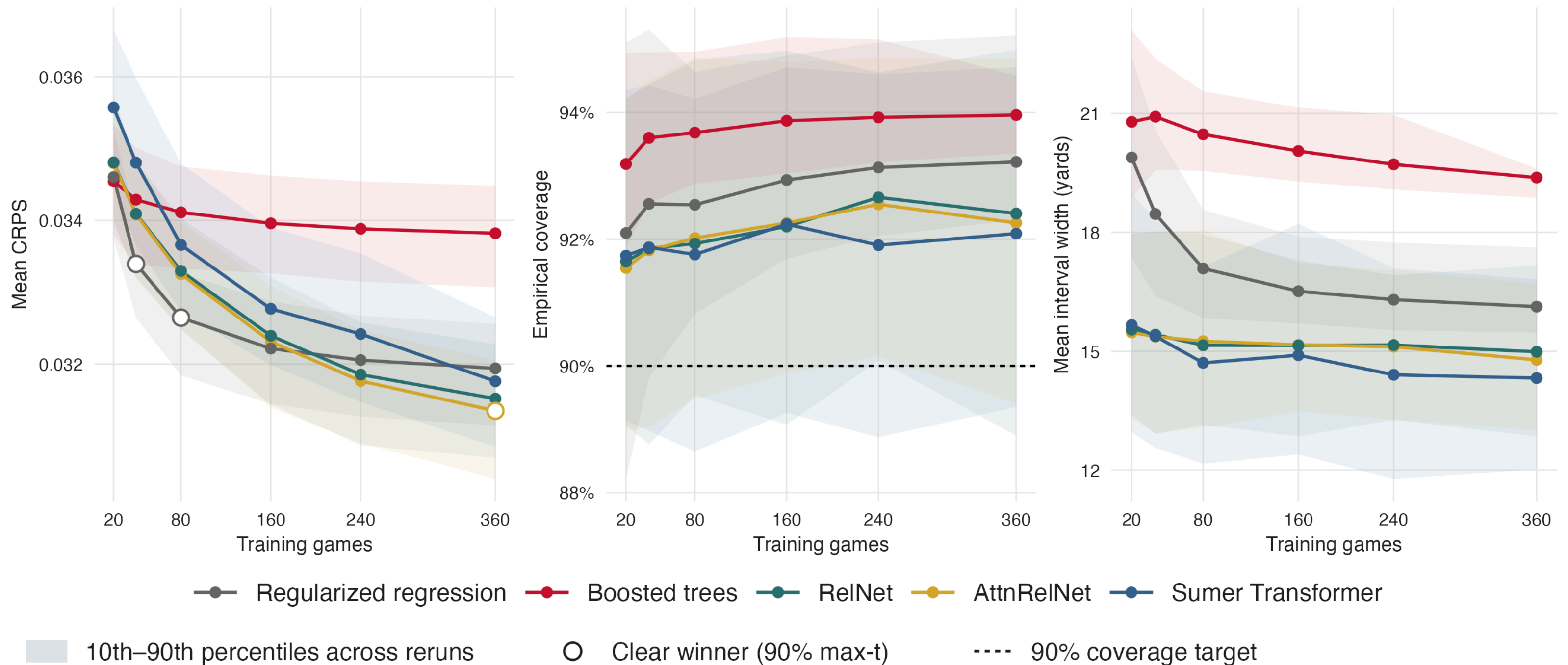
Backup

Additional results

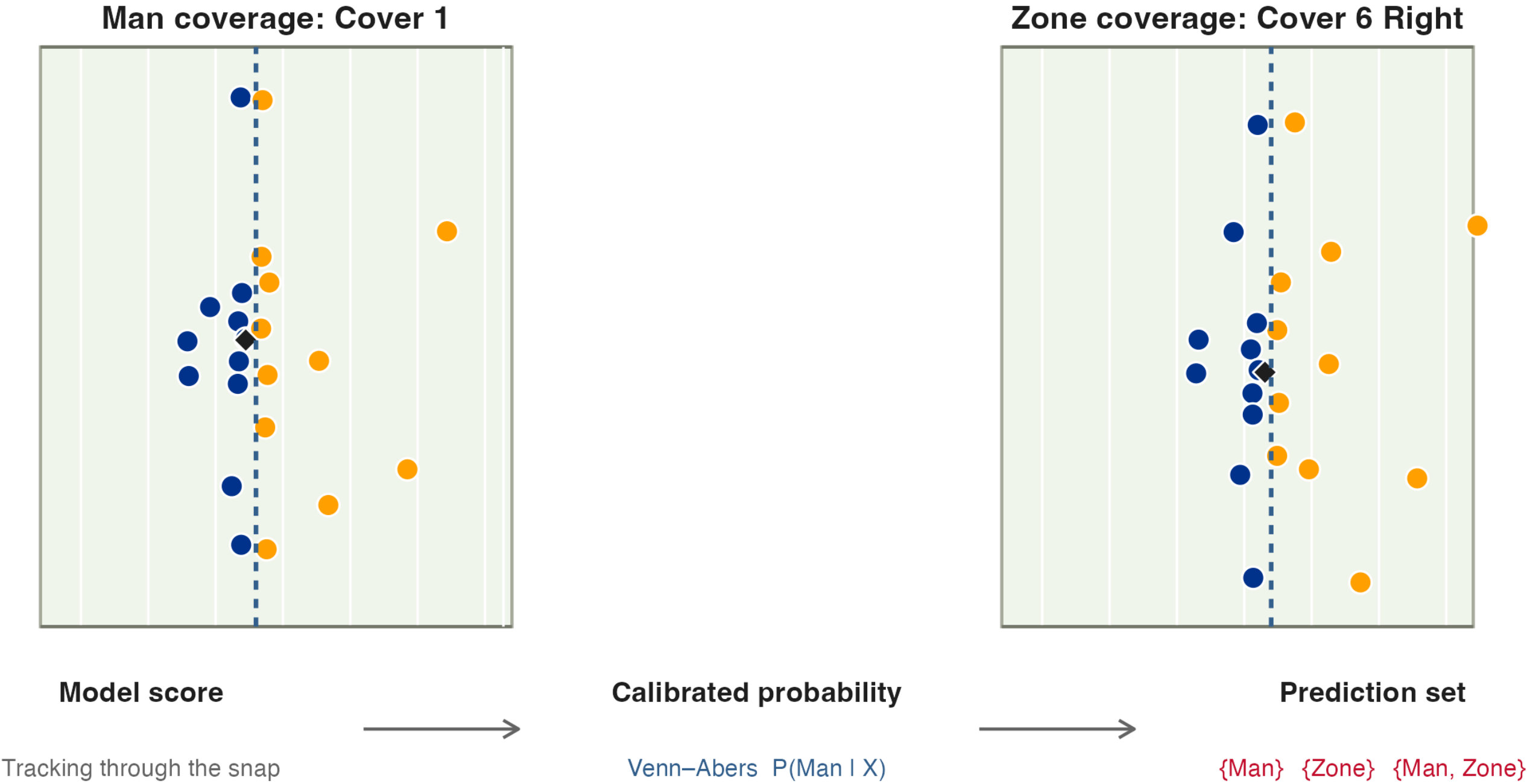
Model inputs

Statistical details

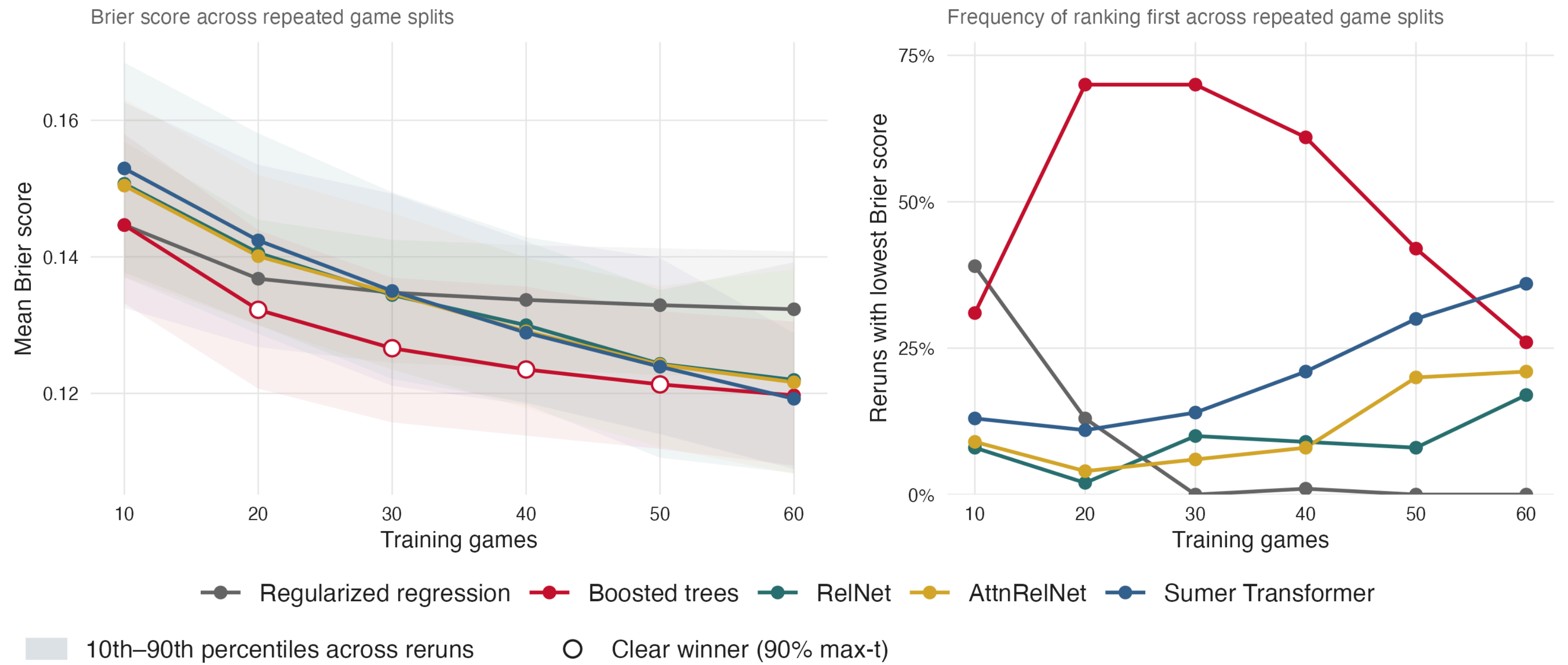
Full BDB2020 learning curves



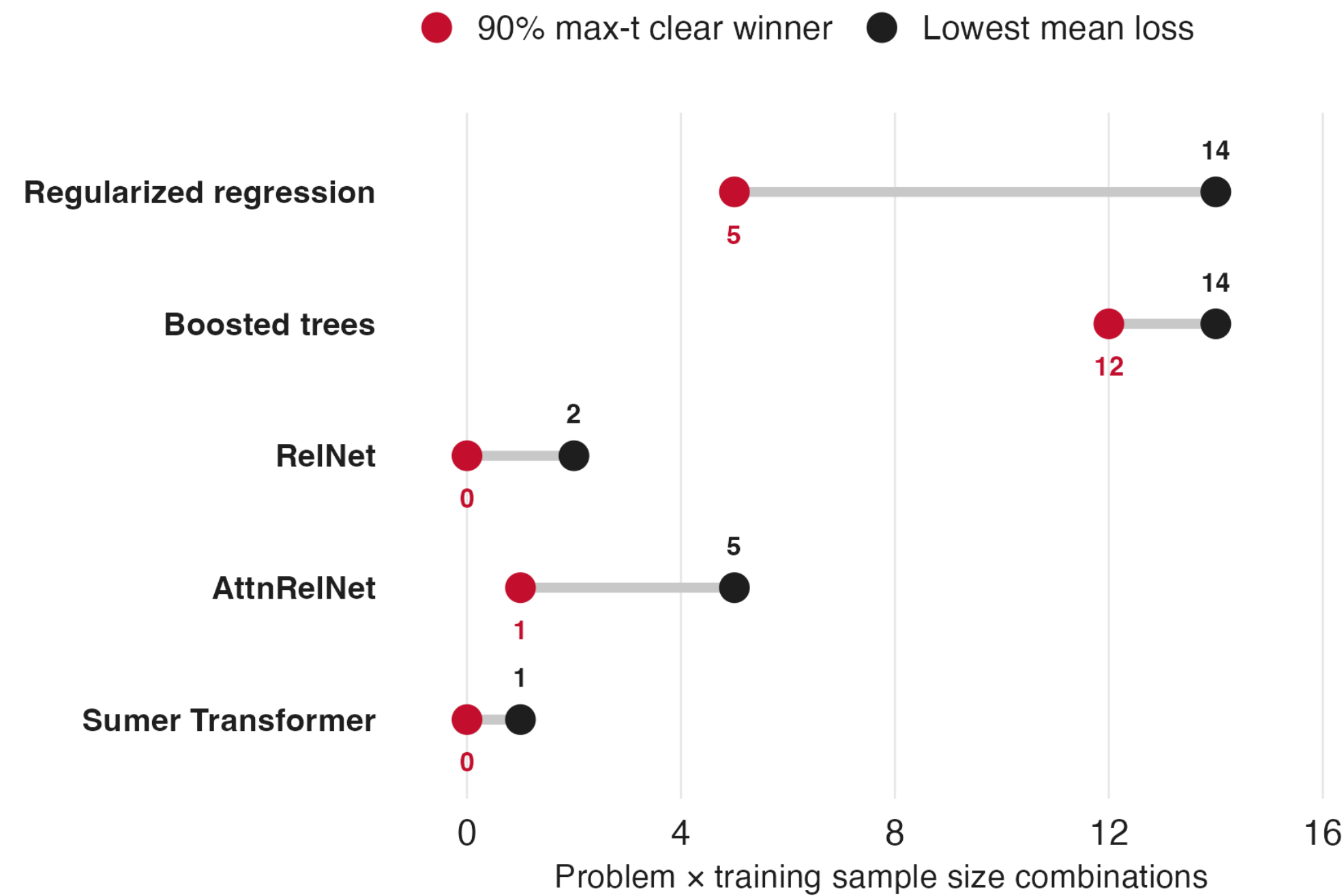
BDB2025: Predicting Man or Zone before the snap



Man/Zone accuracy and ranking stability



Clear winners across problems and sample sizes



18 / 36

problem × training sample size combinations
have a clear winner

1 / 36

RelNet–AttnRelNet 90% intervals exclude zero

An inconclusive difference does not establish equivalence.

Rushing inputs at handoff

One handoff frame. 22 observed players, centered on the rusher and oriented in the offense’s direction.

Play context	Down, yards to go, clock, field position, and score before the play Formation, personnel, and defenders in the box
Tracking	Player positions, velocity, speed, acceleration, direction, and orientation Offense/defense and position group indicators
Engineered features	Position and motion summaries by side Distances between offensive and defensive players and distances from the rusher to other players
Neural input	1 frame × 23 slots × 24 channels The football slot is unused and masked

Man/Zone inputs through the snap

20 frames × 23 objects × 24 channels. Players and football, centered on the football and oriented in the offense’s direction.

Play context	Down, yards to go, clocks, field position, and score before the snap Win probability, expected points, formation, and receiver alignment
Location and motion 6 channels	x, y, horizontal and vertical velocity, speed, acceleration
Angles and object roles 8 channels	Sine and cosine of direction and orientation Indicators for offense, defense, football, and the focal player
Position groups 10 channels	QB, RB, WR, TE, OL, DL, LB, DB, special teams, other

Separate masks track missing players and frames padded on the left. Primary models exclude player and team identity.

Matched model comparisons

Regularized regression vs boosted trees

The same engineered football features. Only the learning algorithm changes.

RelNet vs AttnRelNet

The same tracking, graph, temporal encoder, output head, loss, tuning, seeds, and training protocol.

Mean aggregation vs learned attention weights.

Neural matching tolerances: **5% in parameter count** and **15% in FLOPs**. These constrain computational differences but do not establish equivalence.

Conformal coverage and the sampling unit

Prediction	Calibration unit	Interpretation
Yard interval	One selected play per calibration game	Marginal coverage for one eligible play from a new comparable game
Binary set	One selected example per game and true label	Coverage calibrated separately for each label
Repeated predictions (next step)	Sequence calibration design to be specified	Target: simultaneous coverage across prediction times

The completed analyses require exchangeability at the calibration unit. Coverage is marginal over comparable games, not a guarantee for every play or every subgroup.

Max-t intervals across a problem

Paired loss differences

Comparison j is a model pair trained on the same number of games. Across 100 repetitions, $\bar{d}_j$ is the mean loss difference and s_j its standard error.

Resample complete repetitions

Each of 10,000 bootstrap draws resamples 100 repetition vectors, keeping all 60 comparisons together. Compute the bootstrap mean $\bar{d}_j^*$ and standard error s_j^* , then record

$$M^* = \max_{j=1, \dots, 60} \left| \frac{\bar{d}_j^* - \bar{d}_j}{s_j^*} \right|$$

Simultaneous 90% intervals

Use the 90th percentile of M^* for c , with the ordinary two-sided 90% t critical value as a lower bound.

$$\text{CI}_j = \bar{d}_j \pm c s_j$$

The family covers all model pairs and training sample sizes within one problem.

Example of simultaneous model comparisons

2021: Pass completion, 10 training games

